Table 2: A comparative analysis of IPFS, Filecoin, and Arweave

Feature	IPFS	Filecon	Arweave
Primary function	Content delivery and file sharing	Incentivised, secure storage	Permanent, long-term storage
Incentive layer	No built-in rewards for storage	Yes – miners are rewarded with FIL	Yes – pay miners once with AR
Data type	Mutable – can change over time	Mutable – data can be updated	Immutable – data is stored permanently
Ideal use case	Hosting DApp files, web-sites, media	Large-scale, reliable storage needs	Archiving records, research, NFTs, history
Cost structure	Free, but no storage guar-antees	Pay-as-you-go based on duration and size	One-time payment for forever storage
Protocol type	Protocol for sharing files	Blockchain-powered storage marketplace	Blockweave – a new type of blockchain

economic layer that incentivises long-term storage and reliability. By leveraging blockchain technology, Filecoin allows users to rent out unused hard drive space and earn FIL tokens in return.

How it works: Filecoin creates a decentralised marketplace for storage where users (clients) pay miners to store their data. The protocol ensures trust and integrity through advanced cryptographic proofs.

Step 1. Proof of Replication (PoRep): Miners prove that they have physically stored a unique copy of the client's data.

Step 2. Proof of Spacetime (PoSt): Miners continuously prove that the data is being stored over time.

Step 3. Retrieval miners: Specialised nodes can earn FIL by quickly and efficiently delivering content to users upon request.

Key features

Market-driven storage: An open storage marketplace where providers compete based on cost, speed, and reliability.

Incentivisation model: Encourages miners to offer consistent and long-term data storage backed by economic rewards.

Smart contract integration: Enables programmable interactions with stored data, ideal for Web3 applications and decentralised platforms.

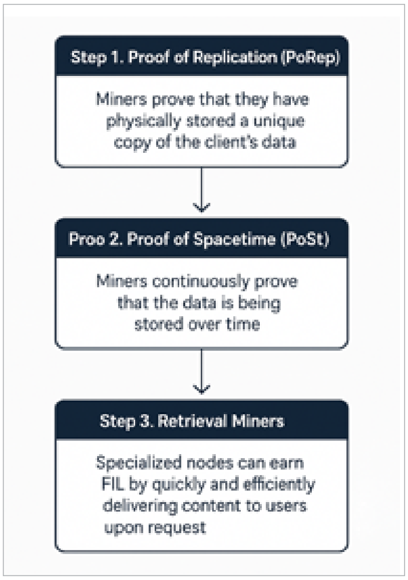

Figure 3: Working of Filecoin

Limitations: Operating a Filecoin node requires significant computational power, storage capacity, and technical expertise, which is a higher barrier for small users. So, due to its infrastructure demands, Filecoin is more suited to organisations and enterprises than casual users or hobbyists.

Arweave

Arweave takes a distinct approach to decentralised storage by emphasising permanent data preservation. Instead of renting space for a period, like traditional or even other decentralised solutions, Arweave operates on a 'pay once, store forever' model. It utilises a blockchain-inspired data structure known as the Blockweave, specifically designed for efficient and permanent archiving of digital content.

How it works: Arweave operates as a decentralised storage network that allows users to store data permanently. When users upload data, they pay a one-time fee using AR tokens. These payments are collected into an endowment fund, which continuously incentivises miners to store and maintain the data over time.

It uses a unique consensus mechanism called Succinct Proof of Random Access (SPoRA), requiring miners to prove they can access random pieces of historical data. This ensures data replication across the network, making the archive tamper-proof and persistently available.

Key features

Permaweb: A decentralised web built on Arweave that hosts permanent web pages, applications, and documents.

True content permanence: Ideal for use cases where immutability is critical—such as NFT metadata, historical archives, academic research, and journalism.

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